Multivariable Calculus Final Project

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1 Construction

1.1 Introduction & Design

For this project, our group chose a cone with radius 30" and height 36" (or $2.5'$ and $3'$, respectively). The equation that we used was

$$\frac{x^2}{225} + \frac{y^2}{225} = \frac{z^2}{900},$$

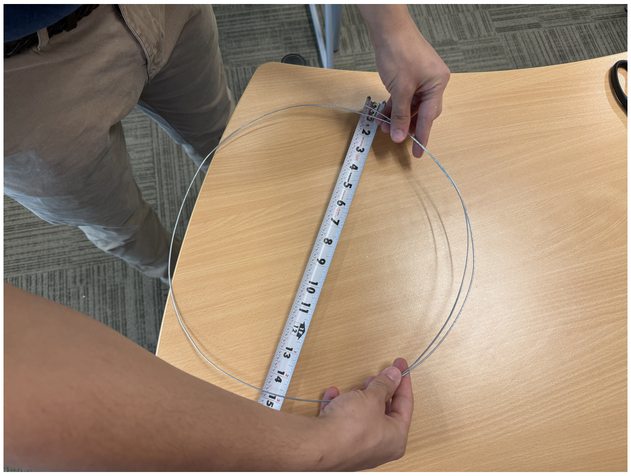
where $x \in [-15, 15]$, $y \in [-15, 15]$, and $z \in [0, 30]$. Our cone was domain-restricted to only positive z values, as it would be physically very difficult to construct a cone that extended into the ground.

1.2 Construction Process

We decided to construct an cone with a peak radius of 15" and a peak height of 30", with the peak radius at the base. To construct the cone, we used wooden sticks to prop up the circles that represented certain radii, and we used metal wire to create those circles. We originally planned to construct a cone with peak radius 30" and a peak height of 36", but Diego pointed out that such a cone would exceed the dimensions of the project area.

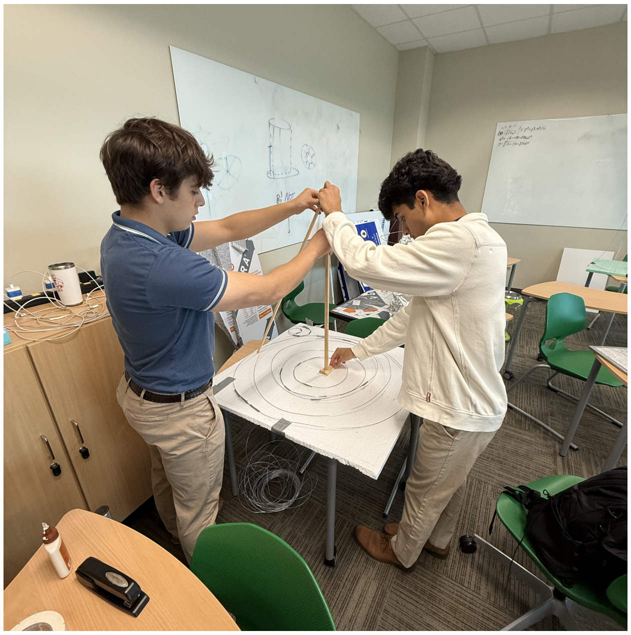
Diego sawing the wooden support sticks.

The construction process consisted of measuring out the radii for the wire circles and placing the wooden sticks on the outer sides of the cone to then place those wires on. The construction process required sawing the wooden sticks, which were at first too long, to the proper lengths to fit on the edges, or hypotenuses, approximately 33.5".



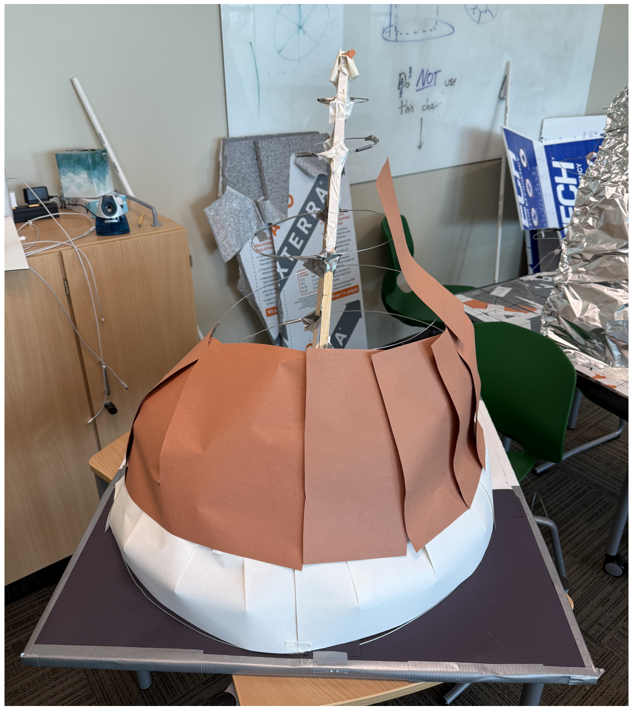
Diego measuring a circle for the cone.

Our construction eventually yielded a finished cone, with five wire circles around the middle and hypotenuse poles. We used wood glue and some, generously, tape to hold everything down. We also asked another group for their extra foam pieces for the platform and borrowed some wooden poles and steel wire. Basically, we survived on communism.



Diego and Keshav working on the wooden sticks.

Afterwards, we started on our theme: ice cream. As the cone was constructed with the peak radius at the base, instead of at the highest z value, we decided to go with a fallen ice cream cone. So we used white construction paper and brown construction paper to wrap one half of the cone, and ripped pieces of green, pink, yellow, blue, and red construction paper for the sprinkles. Our (almost) finished product looked as follows:



The almost finished cone, without sprinkles.

Throughout the project, we all learned different things: Keshav learned how to (hopefully) purchase ice cream for the class and to construct a Riemann sum; Diego learned how to build the project; and I (Michael) learned too many LaTeX formatting methods. In the end, the project was fun. We'll just need to find *something* to do with the final product (maybe Parrie can take it home.)

2 Calculations

2.1 Theoretical Precise Volume

To calculate the theoretical precise value, we will use the triple integral

$$\iiint\limits_Q f(x,y,z)\,dV.$$

To find the bounds, we will use cylindrical coordinates, as Cartesian coordinates tend to end up messy in such calculations. To find the bounds in cylindrical coordinates, we only need to solve for r, as $\theta \in [0, 2\pi]$ and $z \in [0, 30]$, as given by our domain restrictions above.

To find the bounds for r, we first convert the rectangular x and y to cylindrical coordiantes. Following the conversion, our cone can be represented by the equation $\frac{r^2}{225} = \frac{z^2}{900}$, which we can solve to find the upper bound for r:

$$\frac{r^2}{225} = \frac{z^2}{900} \implies r^2 = \frac{225z^2}{900} \implies r = \sqrt{\frac{225z^2}{900}} = \sqrt{\frac{z^2}{4}} = \frac{z}{2}.$$

Thus, the upper bound for r is $\frac{z}{4}$, and because the lower bound for r is 0, we know that the complete bounds for r are $[0, \frac{z}{4}]$. Thus, the complete bounds, for all variables, are:

$$\theta \in [0, 2\pi]\,,\ z \in [0, 30]\,,\ \text{and } r \in \left[0, \frac{z}{2}\right].$$

Therefore, we can plug in the bounds and integrand for the triple integral and evaluate it:

$$V = \int_0^{2\pi}\int_0^{30}\int_0^{\frac{z}{2}} r\,dr\,dz\,d\theta = \int_0^{2\pi}\int_0^{30}\left[\frac{r^2}{2}\right]_0^{\frac{z}{2}} dz\,d\theta = \int_0^{2\pi}\int_0^{30}\frac{z^2}{8}\,dz\,d\theta$$
$$= \int_0^{2\pi}\left[\frac{z^3}{24}\right]_0^{30} d\theta = 1125\int_0^{2\pi} d\theta = 1125\,(2\pi) = 2250\pi \approx 7068.5835.$$

Thus, the theoretical precise volume of our cone is approximately $\boxed{\text{7068.5835 cubic inches}}$ (or approximately 4.0906 cubic feet).

2.2 Theoretical Approximate Volume

To calculate the theoretical approximate value of the equation $\frac{x^2}{225} + \frac{y^2}{225} = (\frac{z^2}{900})$, we must first convert to cylindrical coordinates to find the radius with respect to z, the height. The radius r can be found via the equations:

$$\frac{r^2}{225} = \frac{z^2}{900} \implies r^2 = \frac{z^2}{4} \implies r = \frac{z}{2}.$$

To visualize the Riemann sum, imagine a layer cake with multiple layers that are cylinders of arbitrary h height, and gradually increasing radii. A summation of all the heights of every layer equates to z, or 30. And, note that there are n cylinders. Additionally, at the peak of the inverted cone, or height 30, the radius is 15, and at the minimum, or height 1 (not including height 0), the radius is $\frac{1}{2}$. Now, for the formal calculation.

Let r be the radius of the partition which is a cylinder, h be the height, z be the height of the original cone, and n be the number of partitions that are taken, specifically referring to the height. The radius, with respect to z, is $r = \frac{z}{2}$. The area of each cylinder's circle is $\frac{z^2}{4}\pi$. Therefore, the area of each partition is $\frac{z^2}{4}h\pi$. Finally, let $\Delta z_0 = \frac{30-0}{n}$, and let $z_i = i\Delta z_0$. Inputting all the values into the Riemann sum and taking the limit of it is as follows:

$$V = \sum_{i=1}^{5}[f(z_i)]^2\pi\Delta z_0 = \sum_{i=1}^{5}\frac{z_i^2}{4}\pi\Delta z_0 = \sum_{i=1}^{5}\frac{(i\Delta z_0)^2\pi}{4}\Delta z_0 = \sum_{i=1}^{5}\frac{\pi}{4}i^2\Delta z_0^3 = \sum_{i=1}^{5}\frac{\pi}{4}i^2\frac{30^3}{5^3}$$
$$= \sum_{i=1}^{5}\frac{\pi}{4}i^2\frac{27000}{5^3} = \sum_{i=1}^{5}\frac{6750\pi}{5^3}i^2 = \frac{6750\pi}{125}(1^2) + \frac{6750\pi}{125}(2^2) + \cdots + \frac{6750\pi}{125}(5^2) \approx 9330.5302$$

The theoretical approximate volume of our cone is approximately $\boxed{9330.5302 \text{ cubic inches}}$ (or approximately 5.3996 cubic feet).

2.3 Actual Approximate Volume

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3 Discussion

3.1 Theoretical Precise Volume

Our theoretical precise volume is 7068.5835 cubic inches, and the volume makes sense for the shape that we're planning to construct. Because the volume of the cylinder with $r = 15$" and $z = 30$" can be found via the calculations

$$V = \pi r^2 z = \pi 15^2(30) = \pi 225(30) = 6750\pi,$$

and the final volume is approximately 21205.7504 cubic inches. The volume of our cone is thus contained in the volume of the cylinder, which is logical and appropriate.

3.2 Theoretical Approximate Volume

Our theoretical approximate volume, found with the Riemann sum $\sum_{i=1}^{5} \frac{6750\pi}{125} i^2$ yielded the volume 9330.5302 cubic inches, which makes sense, as we took the cylinders of h height stacked on top of each other, from the base of the cone to the point. Those cylinders are an overestimate, as they were calculated by finding the radius at the base of each cylinder and extending the area of the base to height h, effectively leaving the base area at the next point as well.

To find the percent error of our theoretical approximate volume, we can use the formula for percent error:

$$\text{Percent Error} = \left| \frac{\text{Measured Value} - \text{True Value}}{\text{True Value}} \right| \times 100\%.$$

Our true value would be the value of the integral, 7068.5835, and our measured value would be the value of the Riemann sum, our theoretical approximate value, 9330.5302. The resulting calculations, with those numbers plugged in, would be:

$$\begin{aligned} \text{Percent Error} &= \left| \frac{9330.5302 - 7068.5835}{7068.5835} \right| \times 100 = \left| \frac{2261.9467}{7068.5835} \right| \times 100 \\ &= |0.3199999972| \times 100 \approx 32\% \end{aligned}$$

Thus, our theoretical approximate volume's percent error is approximately 32%.

3.3 Actual Approximate Volume

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